

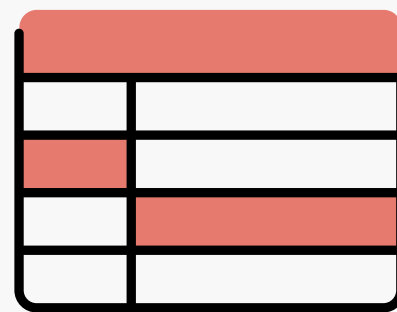
USED CARS PRICE PREDICTION & MARKET ANALYSIS

[GitHub](#)
[Tableau](#)

Automotive Sector | Data Analytics



Data Preparation
(**PYTHON**)



Data Analysis
(**EDA**)



Insights Visualization
(**TABLEAU**)

TEAM MEMBERS

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CONTEXT & PROBLEM STATEMENT

CONTEXT

Millions of used cars are listed online every month, but there is no clear standard for fair pricing.

Pricing is highly inconsistent due to:

- vehicle age
- mileage differences
- subjective seller condition ratings
- regional demand variations
- lack of market transparency

PROBLEM STATEMENT



How can we predict fair used car prices and identify the factors that most strongly influence vehicle pricing?



OBJECTIVE

Identify major pricing drivers → help buyers, sellers, and dealerships make better pricing decisions

27,987

Vehicle Listings Analyzed

\$9,679

Average Vehicle Price

13.37 Years

Average Car Age

DATA ENGINEERING

DATA SOURCE

Craigslist vehicles dataset

ORIGINAL VOLUME

48,472

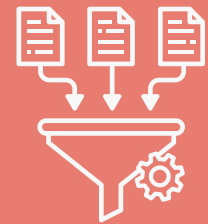


FINAL DATASET

27,987

DATA QUALITY

57.7% retained | 0 nulls | cleaned & standardized



01

Data Collection

Collected ~48,471 vehicle listings from the Craigslist Used Vehicles Dataset (Kaggle)

02

Cleaning & EDA

Handled missing values, removed outliers, standardized variables, and engineered features



03

Analysis & ML

Performed exploratory data analysis, regression analysis, and pricing prediction modeling

04

Visualization

Built interactive Tableau dashboards for market analysis and pricing insights



KPI & METRICS FRAMEWORK

TOTAL LISTINGS

27,987

AVERAGE PRICE

\$9,678.97

AVERAGE CAR AGE

13.37 YEARS

**MOST COMMON
MANUFACTURER**

FORD

HIGH MILEAGE VEHICLES

39%

OLD VEHICLES (>10 YEARS)

63%

⚙️ How are they computed?

- Derived from cleaned dataset (27,987 records)
- Calculated using Python (Pandas)
- Visualized using Tableau Dashboard

KEY INSIGHTS (EDA)

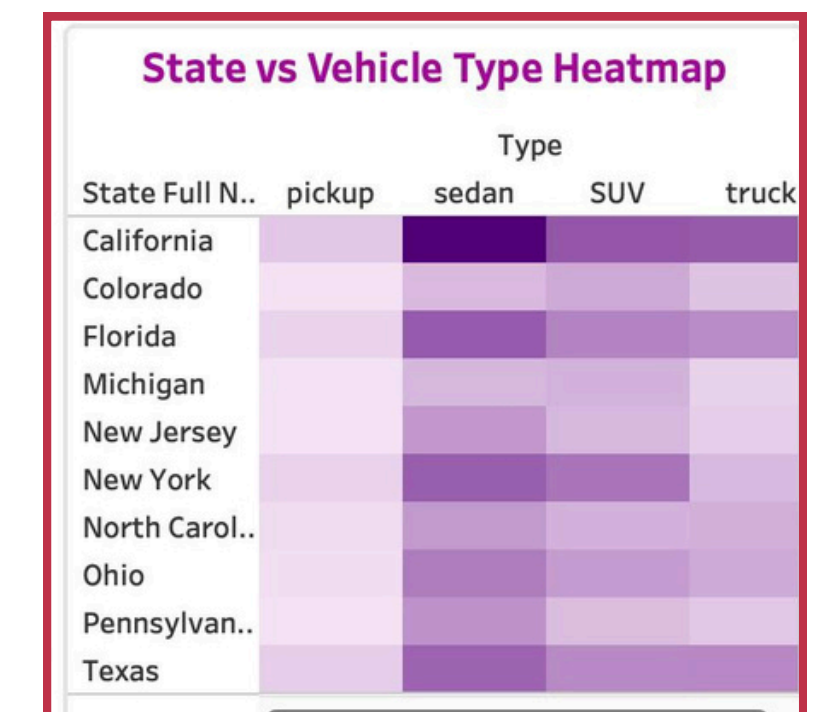
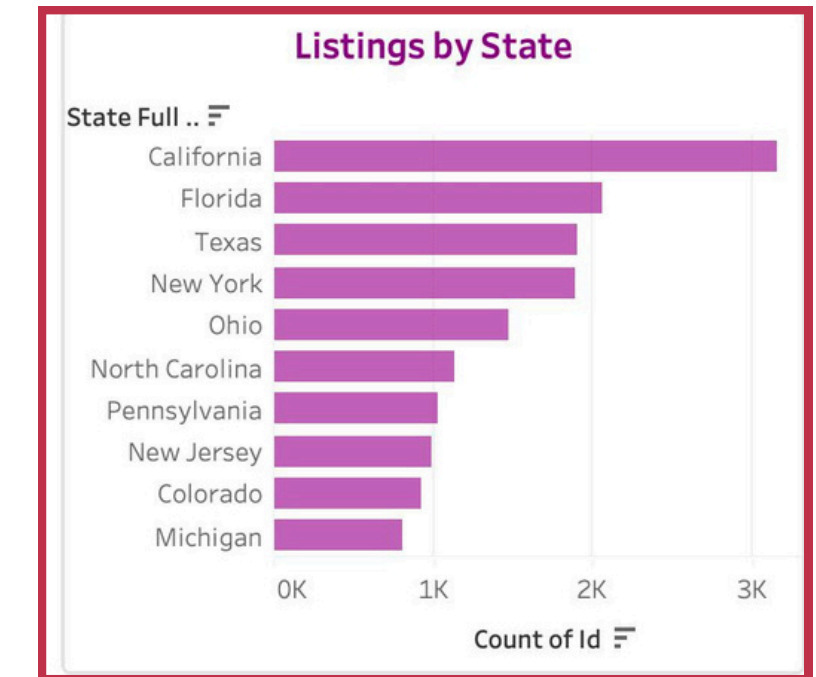
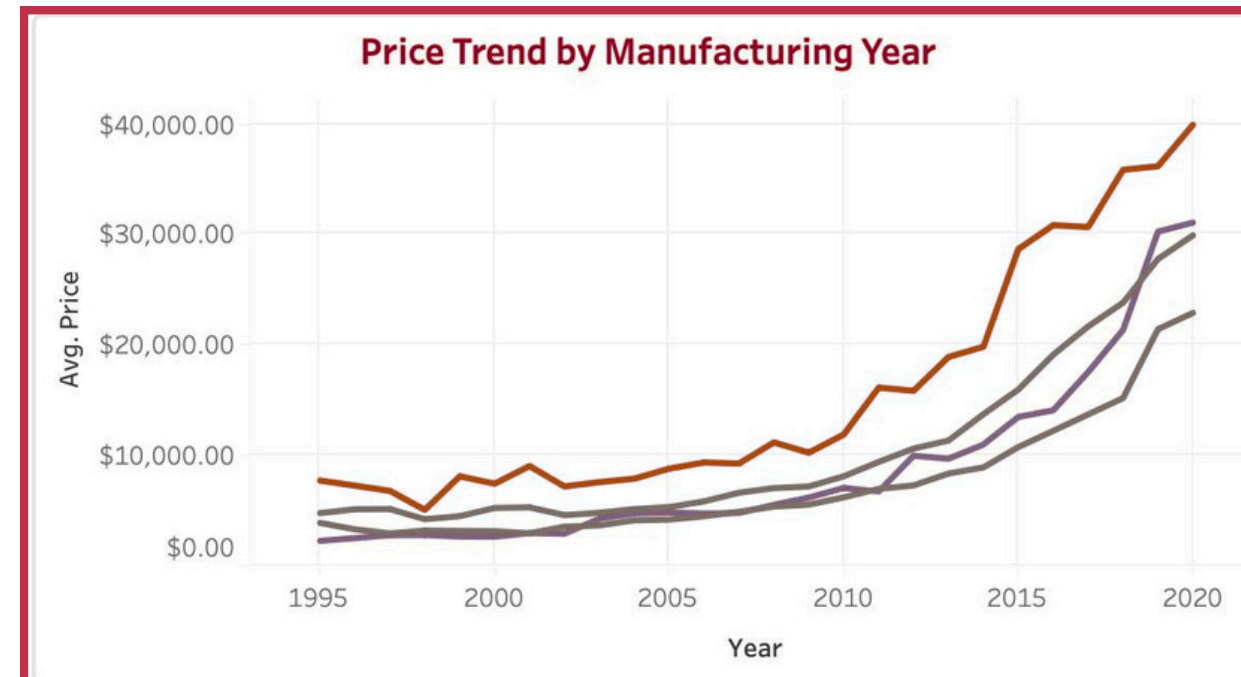
63% of listings are vehicles older than 10 years

39% of vehicles exceed 150,000 miles

Vehicle age is the strongest price predictor

Ford, Chevrolet & Toyota account for nearly 38% of listings

55% of sellers report “excellent” condition → likely listing bias



ADVANCED ANALYSIS



ANALYTICAL DEPTH (WHAT WE DID)

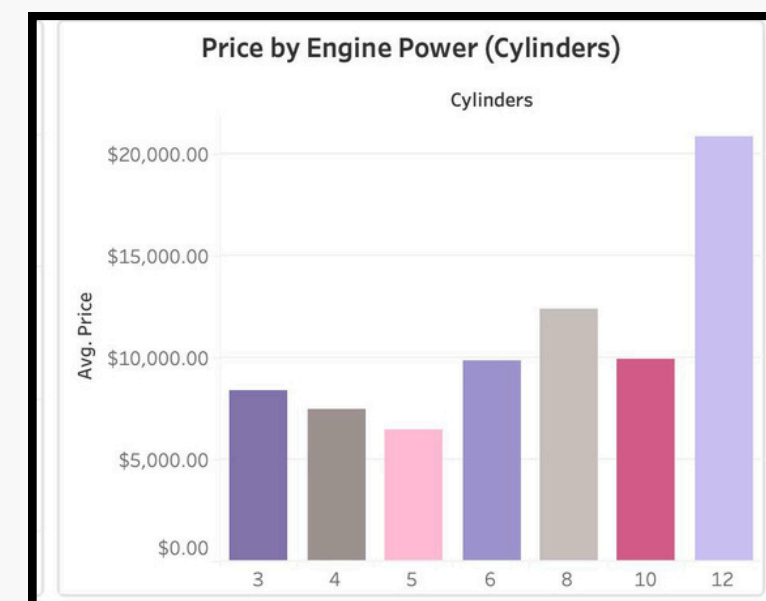
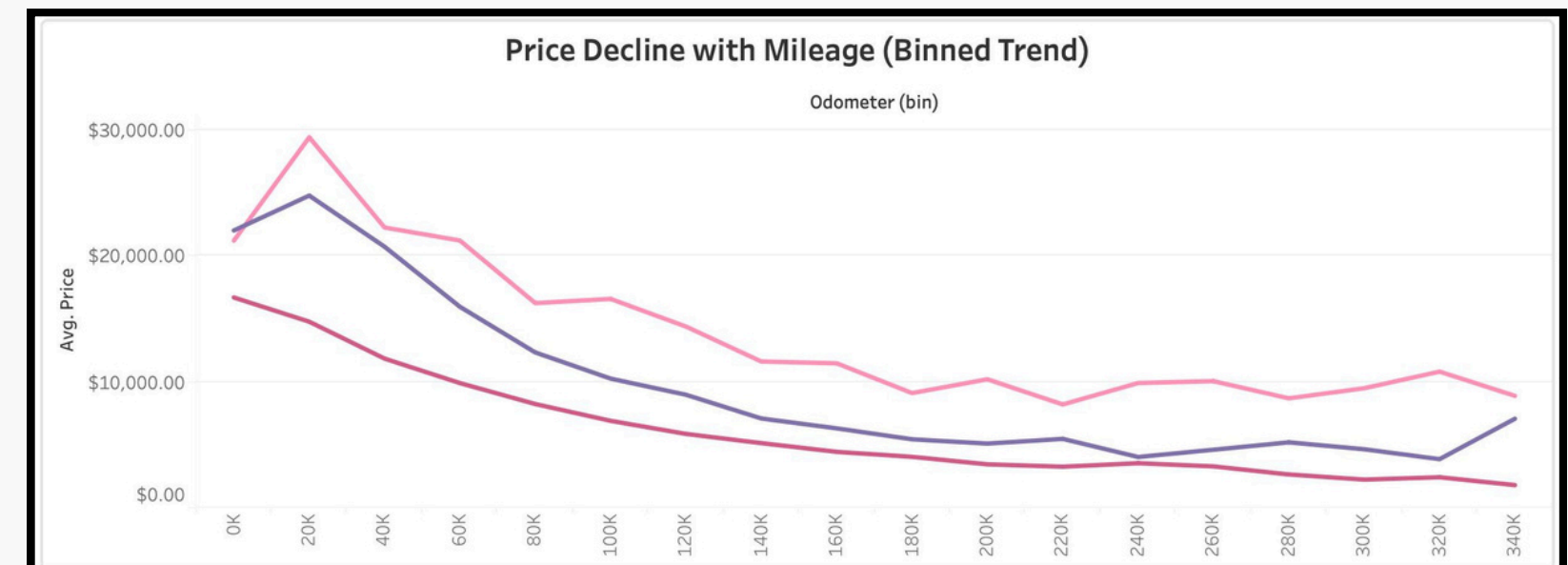
- **Segmentation Analysis** → Vehicle age, mileage, and manufacturer segmentation
- **Regression Analysis** → Linear Regression & Random Forest price prediction
- **Comparative Analysis** → Fuel type, transmission, and condition-based pricing patterns



NEW UNDERSTANDING (BEYOND EDA)

- Vehicles aged 4–8 years show strongest pricing power
- Mileage + age together explain majority of price variation
- Seller-reported condition is unreliable as a standalone pricing signal
- Premium brands command disproportionately higher resale prices

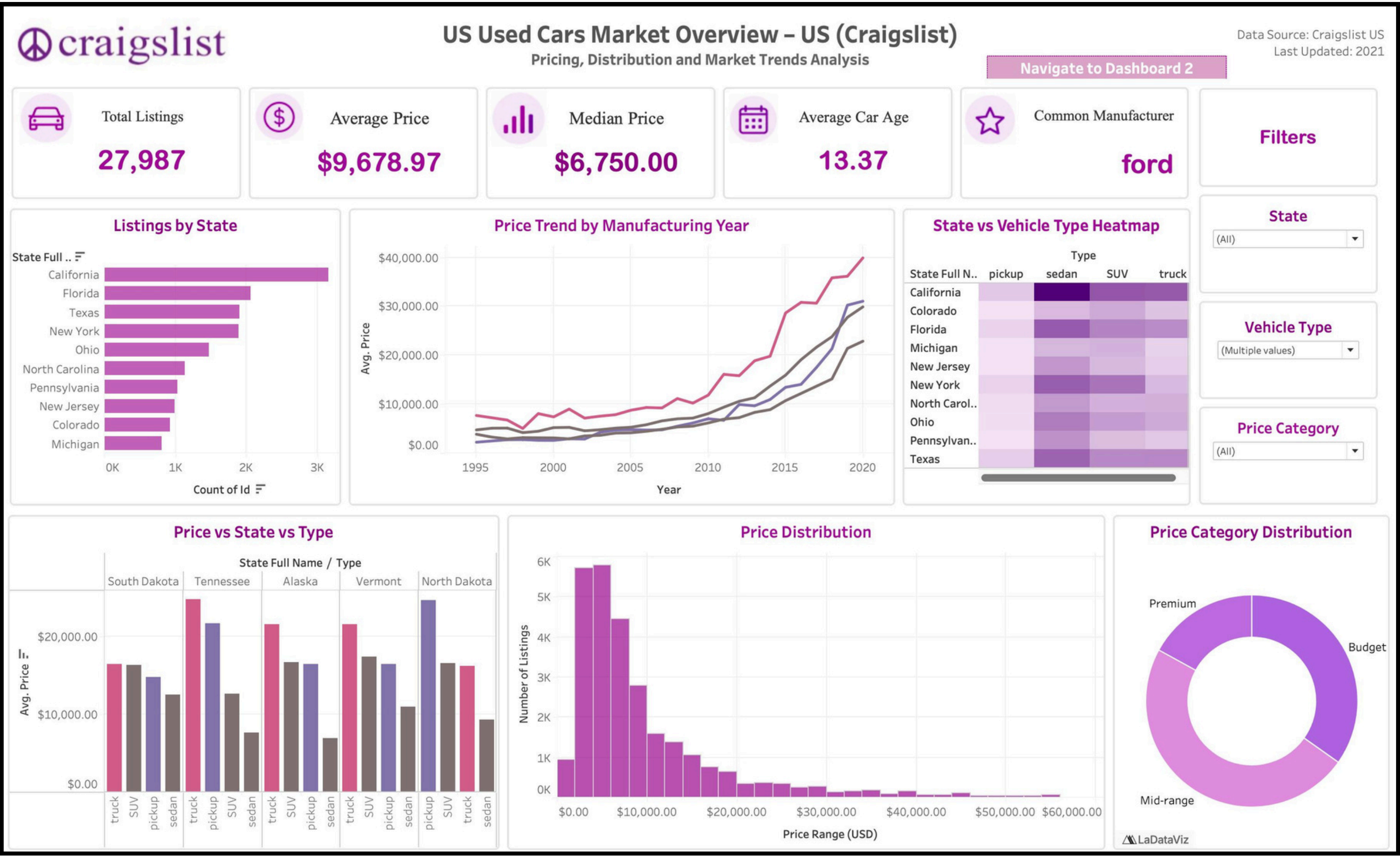
Python (Pandas, NumPy, Scikit-Learn)



DASHBOARD WALKTHROUGH

Link

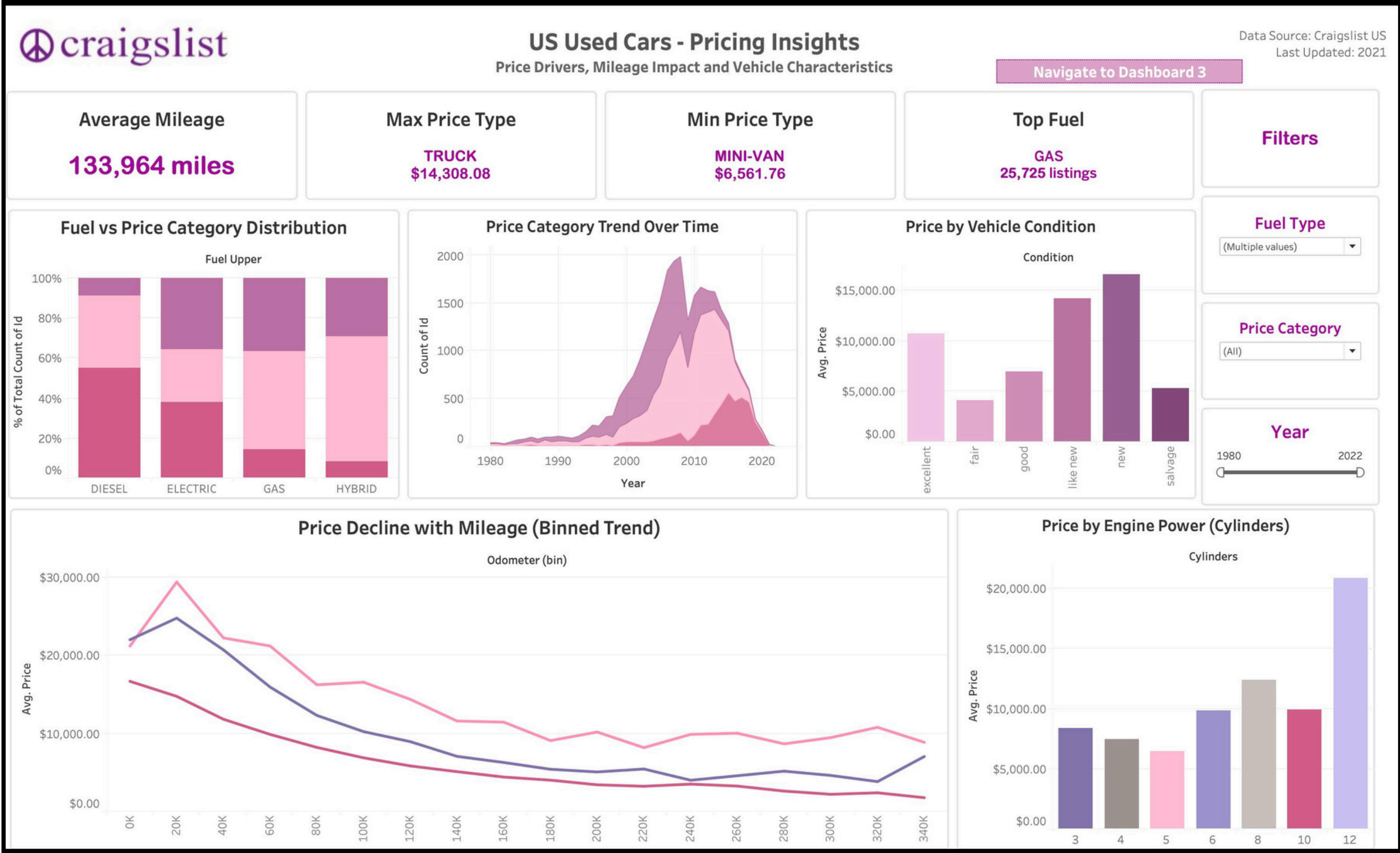
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DASHBOARD WALKTHROUGH

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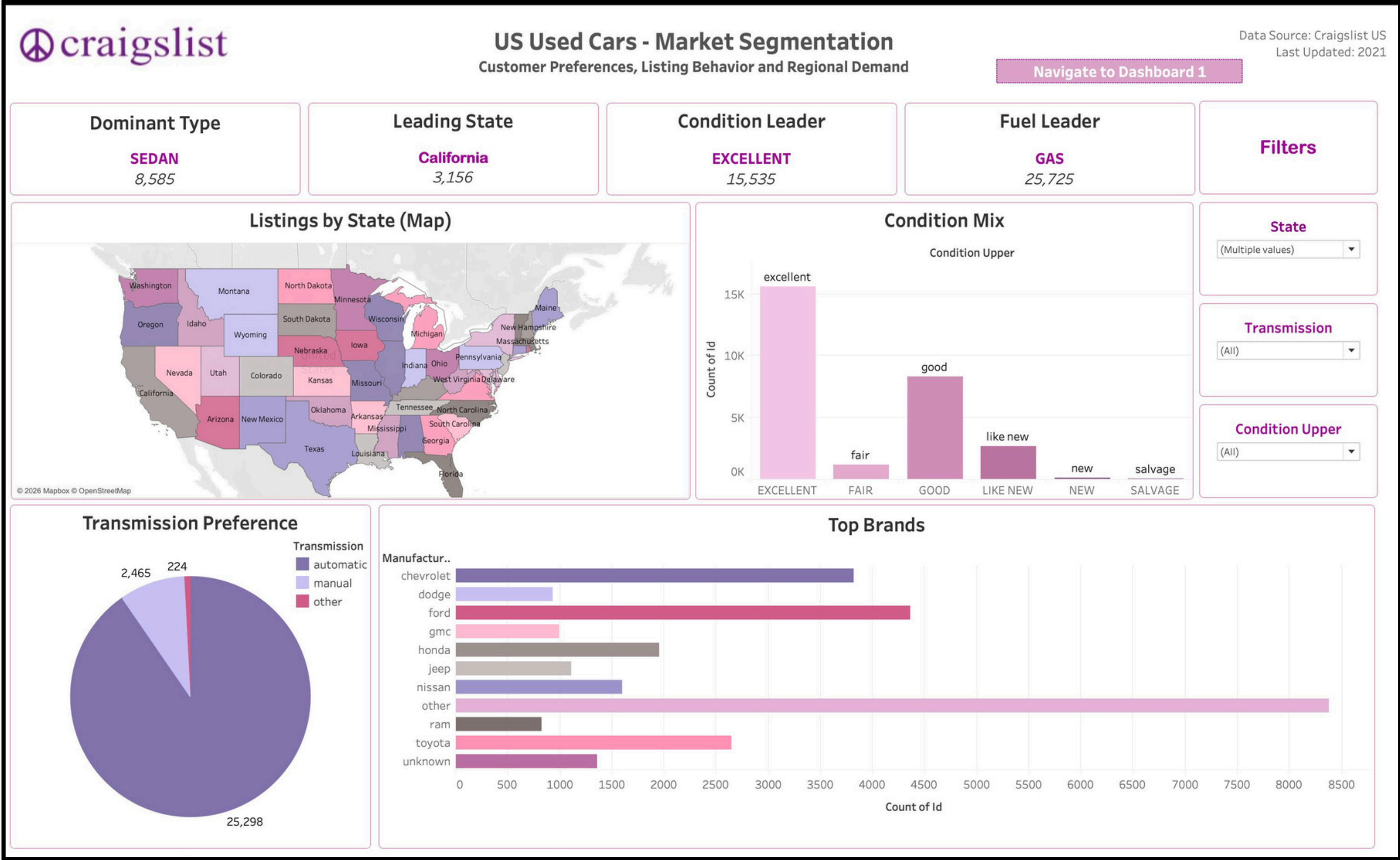
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DASHBOARD WALKTHROUGH

[Link](#)

3



RECOMMENDATIONS

Age & Mileage Benchmarking

- Sellers should compare vehicles by age bracket and mileage before listing
- Reduces underpricing and overpricing

01.

02.

Smart Dealer Acquisition

- Focus on vehicles aged 4–8 years with lower mileage
- Improves pricing power and profit margins

Condition Verification

- Buyers should not rely solely on “excellent” condition labels
- Verify using inspections and mileage benchmarks

03.

04.

Data-Driven Pricing Tools

- Use predictive pricing dashboards for market-based valuation
- Faster and smarter decision-making

IMPACT & VALUE

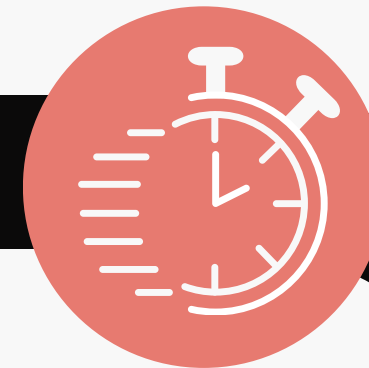


Reduced Pricing Errors

Data-backed benchmarks reduce overpricing and underpricing risk

Better Dealer Margins

Optimized inventory selection improves profitability



Faster Buyer Decisions

Age and mileage benchmarks simplify vehicle evaluation

Data-Driven Decision Making

Dashboard enables faster and more accurate pricing analysis



Impact Snapshot

- **+\$700–\$1,000** seller benefit
- **+\$800–\$1,500** dealership margin potential
- **10%** overpayment avoidance
- **0** additional implementation cost

LIMITATIONS & NEXT STEPS



LIMITATIONS

- Dataset contains Spring-only listings → no seasonal trend analysis
- Vehicle condition is self-reported and potentially biased
- No vehicle history data (accidents/service records)
- Electric vehicle sample size is too small for reliable analysis



FUTURE ROADMAP

- Real-time price prediction system
- Full-year dataset expansion
- Advanced ML model deployment
- Fraud/anomaly detection for overpriced listings